

StoryTime MVP PRD v0.4 — Hardened Scope

1. Product Definition

StoryTime is a parent-mediated mobile app that turns a remote family call into a short, guided, AI-assisted story chapter and saves that chapter as a private replayable family memory.

The product is not a general video-calling app, not a child-facing AI chatbot, and not a public story platform. The atomic unit of value is:

One completed story chapter that a child wants to replay and a parent feels safe preserving.

2. Launch Segment

The MVP targets high-connectivity families where at least one adult is regularly away from the child.

Initial target:

- English-speaking families
- Children aged 6–10
- Travelling parents
- Dual-household or separated parents
- Long-distance grandparents only where a parent/guardian manages the account
- Modern iOS/Android devices
- Stable 4G/5G or Wi-Fi environment

MVP does not target low-bandwidth migrant-family use cases as the primary wedge. Those remain strategically important but require an audio-first version or degraded-network product path.

3. Core Product Promise

StoryTime helps a remote parent and child create a magical 10–15 minute story together, see fast AI-generated scenes, and save the whole story event as a private memory chapter.

The emotional promise:

Dad is away, but the child can replay the moment they laughed, invented a dragon, and finished the adventure together.

4. MVP Product Unit

The MVP product unit is a completed chapter.

A completed chapter includes:

- adult-authenticated session start
- verified child-side adult handoff
- story setup
- turn-based narration
- AI-supported story beats
- generated visual scenes
- recording of parent/child voices and video tracks
- chapter processing
- private replay in the family library

Campaigns may exist as lightweight metadata, but the first release must optimise for chapter completion and replay before complex long-running campaign mechanics.

5. Non-Negotiable Release Gates

The MVP cannot launch unless the following gates are met:

1. Verifiable parental consent is completed before any child voice/video collection.
2. Live call connection is stable enough for beta use.
3. Recording boundaries are clear and visible to adults.
4. Every chapter has a recoverable recording state.
5. Raw tracks and AI/UI events can be composed into a replayable chapter.
6. Unsafe AI outputs are blocked or replaced with safe fallbacks.
7. Parent deletion controls work end-to-end.
8. Cost per completed chapter is measurable.

6. MVP User Flow

First-use setup

Adult creates account

Adult verifies identity / parental consent

Adult creates child profile

Adult invites approved co-parent or family adult

System confirms both adults are approved for the child profile

Story call

Remote parent starts Story Call

Child-side adult receives Adventure Call Alert

Child-side adult authenticates

Child-side adult sees remote parent

Child-side adult confirms handoff

Recording notice appears

Child-safe mode starts
Story setup begins
Recording starts at story setup

Story chapter

Child and remote adult choose story ingredients
Story seed locks
Baton alternates between speakers
Active speaker talks
Speaker taps pass
System transcribes the turn
Story Director creates concise story beat
Image generator creates fast visual scene
Scene appears on both screens
Timeline event is logged
Chapter continues until checkpoint or ending

Chapter end

Adult chooses pause, end, or create ending
Recording stops
Raw tracks are stored
Timeline ledger is finalised
Async compositor generates replay
Chapter appears in Memory Vault

7. Functional Requirements

Adult account and consent

The system must require an adult account before any child session can start.

The system must store:

- adult account ID
- consent status
- consent method
- consent timestamp
- child profile association
- approved adults
- privacy notice version
- terms version

No child-facing session may start if consent is missing, expired, withdrawn, or inconsistent with the product's current data practices.

Child profile

A child profile must be controlled by an adult.

The child profile may include:

- display name
- age band
- preferred story styles
- approved adults
- replay permission
- deletion status

The child must not have an independent public account.

Story setup

The MVP must include five setup categories:

- character
- setting
- problem
- tone
- visual format

The system must lock the story seed before live story turns begin.

Baton storytelling

Only the active baton holder's speech is processed for AI story generation.

The system must preserve open live audio between participants but process story turns only when the active speaker submits a turn.

AI output

The Story Director must generate:

- transcript
- short story beat
- child-safe caption
- image prompt
- next-turn prompt

The AI must not hijack the story, correct the child harshly, introduce adult themes, or continue after a blocked safety event without safe fallback handling.

Replay

The completed chapter must be replayable inside the parent dashboard.

Replay must show:

- story title
- chapter number
- duration
- generated scenes
- parent/child video composition
- chapter status
- storage used
- deletion control

8. Technical Architecture

Recommended MVP stack

Mobile app: React Native with TypeScript

Video: LiveKit

Realtime state: Convex or equivalent realtime backend

Storage: Cloudflare R2

AI STT: low-latency transcription provider

Story Director: structured-output LLM

Image generation: low-latency image provider

Billing: RevenueCat later in MVP

Async composition: background FFmpeg/media worker

Mobile architecture rule

Expo Managed Workflow is not automatically accepted for MVP.

The team must run a media reliability spike before locking the stack. If LiveKit video reliability is poor in Expo Managed Workflow, the product must move to Bare React Native or a native Swift/Kotlin media layer.

Recording architecture

MVP must use raw-track capture plus asynchronous composition.

The live session records:

- adult video track
- child-side video track
- adult audio track
- child-side audio track

- generated image assets
- story metadata
- timestamped UI/story events

The final replay is generated after the session using the raw tracks and timeline ledger.

RoomComposite is not the default MVP recording path because of compute cost and scaling risk.

Chapter Composition Ledger

Every important story and UI event must be timestamped.

Required events:

- recording started
- story setup started
- setup option shown
- setup choice selected
- story seed locked
- turn started
- turn submitted
- transcript ready
- story beat ready
- image prompt ready
- image ready
- baton passed
- checkpoint shown
- ending started
- session ended
- recording stopped
- composition started
- replay ready
- replay failed

The async compositor must use this ledger to align video, audio, generated scenes, captions, and baton states.

9. Safety and Privacy Requirements

StoryTime processes highly sensitive family media.

The system must provide:

- explicit recording notice
- verifiable parental consent

- no public child profiles
- no open social graph
- no ads
- no behavioural targeting
- no model training on family media by default
- parent deletion controls
- retention limits
- processor inventory
- audit logs for deletion and export actions

The consent screen must clearly state that StoryTime records and stores child voice/video, parent voice/video, transcripts, generated story content, and generated images for private replay.

10. AI Safety Requirements

The system must moderate:

- child/adult transcript
- story beat
- image prompt
- generated image where technically feasible
- child profile inputs
- user-entered names or custom story settings

Safety fallback behaviour:

- replace unsafe story beat with safe alternative
- replace unsafe image with safe placeholder
- notify adult only where needed
- continue the story without exposing technical or frightening language to the child

11. Monetisation and Usage Controls

MVP beta should not rely only on storage caps.

The product must track:

- storage used
- number of generated images
- transcription minutes
- LLM calls
- image-generation calls
- replay bandwidth
- composition cost

- cost per completed chapter

Free beta should include limited weekly Adventure Credits and limited Memory Vault storage.

Paid tiers later unlock:

- more storage
- more Adventure Credits
- standard MP4 download
- premium remastering
- longer archive retention
- cloud backup
- storybook export

12. MVP Success Metrics

North-star metric:

- completed story chapters per active family per week

Activation metrics:

- invited families completing first chapter within 7 days
- first-call connection success
- consent completion rate
- child handoff completion rate

Core value metrics:

- chapter completion rate
- chapter replay within 7 days
- second chapter within 30 days
- average turns per chapter

Reliability metrics:

- call connect success rate
- P95 baton-to-image latency
- recording start success
- replay composition success
- chapter recovery success after network drop

Trust metrics:

- deletion request completion
- blocked unsafe output rate

- consent failure rate
- parent support tickets per 100 chapters

Unit economics:

- cost per completed chapter
- inference cost per chapter
- storage cost per active family
- replay bandwidth cost
- gross margin per paid family

13. MVP Non-Goals

The MVP must not include:

- public child accounts
- public sharing
- stranger discovery
- child-facing purchases
- ads
- voice cloning
- open-ended AI chat
- complex 3D worlds
- classroom mode
- marketplace
- hardware integration
- fully featured campaign universe
- bulk export archive
- printed storybooks
- low-bandwidth migrant-family primary positioning

14. Product Principle

StoryTime must make the adult and child feel like they created the story together.

AI is stage crew, not the star.

The saved memory is not just the story output. It is the shared moment: the voices, laughter, choices, reactions, and the child's imagination unfolding with someone they love.